

Problem I

Problem II

Problem (Problem II.1): Let V and W be $\mathbb{K}$ -TVSs, and let $T: V \rightarrow W$ be a continuous linear map. Show that if $(x_i)_i$ is a Cauchy net in V , then $(Tx_i)_i$ is a Cauchy net in W .

Solution: Let U be a neighborhood of 0 in W . Then, the inverse image $T^{-1}(U)$ is an open neighborhood of 0 in V . There is some index k such that for all $i, j \in I$ with $k \leq i$ and $k \leq j$, we have that $x_i - x_j \in T^{-1}(U)$. Yet, this means that $Tx_i - Tx_j \in U$ for all $i, j \in I$ with $k \leq i$ and $k \leq j$. This gives the desired result.

Problem (Problem II.2): Let $(V, \mathcal{T}(\mathcal{N}))$ be a LCTVS, and let $(x_i)_i$ be a net in V . Let $\bar{x} \in V$ be some element. Show that the following are equivalent:

- (i) the net $(x_i)_i$ converges to $\bar{x}$;
- (ii) for all $\rho \in \mathcal{N}$ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i \in I$ with $k \leq i$, one has $\rho(x_i - \bar{x}) \leq \varepsilon$;
- (iii) for all continuous seminorms ρ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i \in I$ with $k \leq i$, one has $\rho(x_i - \bar{x}) \leq \varepsilon$.

Solution: Since translation by a vector is a homeomorphism, we will assume that $\bar{x} = 0$ and prove the results accordingly.

Suppose $(x_i)_i \rightarrow 0$. If ρ is a defining seminorm, then we observe that the subbasic open set

$$U(0, \rho, \varepsilon) = \{y \in V : \rho(y) < \varepsilon\}$$

is an open neighborhood of 0. In particular, this means there is some $k \in I$ such that for all $i \in I$ with $k \leq i$, one has $x_i \in U(0, \rho, \varepsilon)$, or that $\rho(x_i) < \varepsilon$. This gives (ii).

To show (iii) from (ii), one observes that if ρ is a continuous seminorm, then there are defining seminorms $\rho_1, \dots, \rho_n$ and a constant C such that

$$\rho(x) \leq C \max\{\rho_1(x), \dots, \rho_n(x)\}.$$

Given $\varepsilon > 0$, for each $j = 1, \dots, n$ we select $k_j \in I$ such that for all $i_j \in I$ satisfying $k_j \leq i_j$, one has $\rho(x_{i_j}) < \varepsilon/C$. There is some k such that $k_j \leq k$ for each $j = 1, \dots, n$. Then, for any $i \in I$ with $k \leq i$, one has $\rho(x_i) \leq \varepsilon$, yielding (iii).

Finally, assuming (iii) holds, we observe that for any neighborhood U of 0, there is some continuous seminorm ρ and some $\varepsilon > 0$ such that

$$U(0, \rho, \varepsilon) = \{y \in V : \rho(y) < \varepsilon\} \subseteq U.$$

In particular, this means that there exists some $k \in I$ such that for all $i \in I$ with $k \leq i$, one has $\rho(x_i) \leq \varepsilon/2 < \varepsilon$ by the assumption of (iii). In particular, this means for all $k \leq i$, one has $x_i \in U(0, \rho, \varepsilon) \subseteq U$. Thus, the net $(x_i)_i$ converges to x .

Problem (Problem II.3): Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space, and let $(x_i)_i$ be a net in V . Show that the following are equivalent:

- (i) the net $(x_i)_i$ is Cauchy;
- (ii) for all $\rho \in \mathcal{N}$ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i, j \in I$ with $k \leq i$ and $k \leq j$, one has $\rho(x_i - x_j) \leq \varepsilon$;
- (iii) for all continuous seminorms ρ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i, j \in I$ with $k \leq i$

and $k \leq j$, one has $\rho(x_i - x_j) \leq \varepsilon$.

Problem III

Problem (Problem III.1): We say a neighborhood U of the origin in a topological vector space $(V, \mathcal{T})$ is balanced if and only if for all $x \in U$ and all $\lambda \in \mathbb{K}$ with $|\lambda| \leq 1$, we have $\lambda x \in U$. Show that for all open neighborhoods $U \in \mathcal{O}_V(0)$, there exists $B \in \mathcal{O}_V(0)$ with $B \subseteq U$ and B balanced.

Solution: Let U be an open neighborhood of 0. Then, since the scalar multiplication map $m: \mathbb{K} \times V \rightarrow V$ is continuous, it follows that $m^{-1}(U)$ is an open subset of $\mathbb{K} \times V$. Therefore, there is a basic open subset $U(0, \varepsilon) \times W \subseteq m^{-1}(U)$, where

$$U(0, \varepsilon) = \{z \in \mathbb{K} : |z| < \varepsilon\}.$$

It follows that

$$m(U(0, \varepsilon) \times W) = \bigcup_{|z| < \varepsilon} zW$$

is a balanced open subset of U containing 0. This yields our desired result.

Problem (Problem III.2): Show that, in a topological vector space, if two subsets A and B are bounded, then $A \cup B$ is bounded.

Solution: Let $U \in \mathcal{O}_V(0)$ be a neighborhood of 0. Then, there is a balanced subset $W \in \mathcal{O}_V(0)$ with $W \subseteq U$. There is ε_1 such that $\varepsilon_1 A \subseteq W$ and $\varepsilon_2 B \subseteq W$, so taking $\varepsilon = \min(\varepsilon_1, \varepsilon_2)$ yields that $\varepsilon A \subseteq W$ and $\varepsilon B \subseteq W$.